

Sachin Loddiya Karthik

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Skills

- Python | R | SQL | Java | C++ | Machine Learning | PyTorch | TensorFlow | Scikit-Learn | XGBoost | OpenCV | MLflow | NLTK
- LLMs | LangChain | FAISS | OpenAI API | Gemini | NLP | Conversational AI | RAG Systems | Text Embeddings | Generative AI
- Azure | AWS | GCP | Databricks | Apache Spark | Kafka | Airflow | Docker | Kubernetes | CI/CD | Jenkins | Docker | CI/CD
- PostgreSQL | MongoDB | Power BI | Tableau | Streamlit | FastAPI | Git | Jupyter | Statistical Analysis

Experience

- | | | | |
|---|--|-----------------------|--------------------------|
| AI Engineer | <u>Goodie Bag</u> | Remote, USA | 06/2025 – Present |
| <ul style="list-style-type: none">• Developed an AI-powered Customer Marketing Agent to generate branded copy across channels (social media, email, website, in-app content, and physical ads), ensuring consistent brand voice.• Orchestrated RAG (Retrieval-Augmented Generation) pipelines using brand guidelines, historical campaign data, and tone-of-voice documents to improve message consistency.• Designed and deployed a stateful conversational AI with FastAPI to handle 7+ customer support use cases, reducing manual ticket handling by 30%.• Engineered a high-performance, non-blocking API processing 50000+ daily transactions for real-time refunds and ticket creation.• Built a PostHog-powered analytics dashboard by writing custom HogQL queries to track active users across multiple cities, enabling dynamic date filtering and actionable insights for growth teams. | | | |
| Data Scientist | <u>WMU</u> | Kalamazoo, USA | 01/2025 – 04/2025 |
| <ul style="list-style-type: none">• Created an automated class scheduling optimizer using Google OR-Tools, reducing manual scheduling effort by 85% for 200+ course assignments• Built a Streamlit interface for real-time CSV ingestion/validation, minimizing data errors by 90%.• Applied operations research and constraint optimization to generate 100% conflict-free schedules for 5 academic departments, accommodating 20+ instructor preferences and 15+ classroom resource constraints | | | |
| Data Scientist | <u>Green Expectations LLC</u> | Remote, USA | 01/2024 – 04/2024 |
| <ul style="list-style-type: none">• Formulated a rule-based Home Sustainability Scoring model leveraging 1,000+ data points, enhancing recommendation accuracy by 35%.• Refined NLP chatbot workflows, reducing latency by 40% and improving user engagement by 25%.• Directed feature engineering for sustainability predictions across varied user profiles, improving model accuracy by 15%. | | | |
| Data Engineer | <u>Accenture Solutions</u> | Remote, India | 07/2021 – 07/2023 |
| <ul style="list-style-type: none">• Architected ETL pipelines using Azure Data Factory and Databricks, improving data processing efficiency by 40% and accuracy by 25%.• Enhanced batch transformation workflows on Data Lake Gen2, halving processing times while maintaining schema consistency across diverse datasets.• Constructed Power BI dashboards and integrated CI/CD pipelines, accelerating delivery speed by 30%. | | | |
| Data Engineer | <u>Claritrics India Pvt Ltd</u> | Remote, India | 11/2020 – 05/2021 |
| <ul style="list-style-type: none">• Implemented OCR-based document processing systems via OpenCV and Python, boosting text extraction accuracy by 30%• Streamlined recurring workflows by automating manual processes, reducing processing time by 40% and boosting team efficiency. | | | |

Education

- | | | | |
|--|---|----------------|--------------------------|
| Master of Science | <u>WESTERN MICHIGAN UNIVERSITY</u> | MI, USA | 08/2023 – 04/2025 |
| <ul style="list-style-type: none">• Data Science | | | GPA: 3.86 |

Project

Ask My PDF – LLM-Powered Document Q&A System (LangChain, FAISS, BM25, PyMuPDF):

- Engineered a scalable, multilingual PDF Q&A system that combines LLMs with hybrid retrieval (FAISS + BM25) to deliver context-aware answers over long, unstructured documents.
- Launched a Streamlit-based web app supporting multi-PDF uploads and memory-aware chat sessions, delivering real-time AI-powered responses that reduced user query time by 40%

Enhanced ATS Resume Expert Pro (Streamlit, Gemini Pro, PyMuPDF, scikit-learn, Plotly):

- Engineered a scalable, multilingual PDF Q&A system that combines LLMs with hybrid retrieval (FAISS + BM25) to deliver context-aware answers over long, unstructured documents.
- Integrated PDF/DOCX parsing, TF-IDF + cosine similarity, and skills gap detection to provide actionable insights, boosting resume relevance scores by up to 30% for users.

Certification

CSM (Certified Scrum Master)